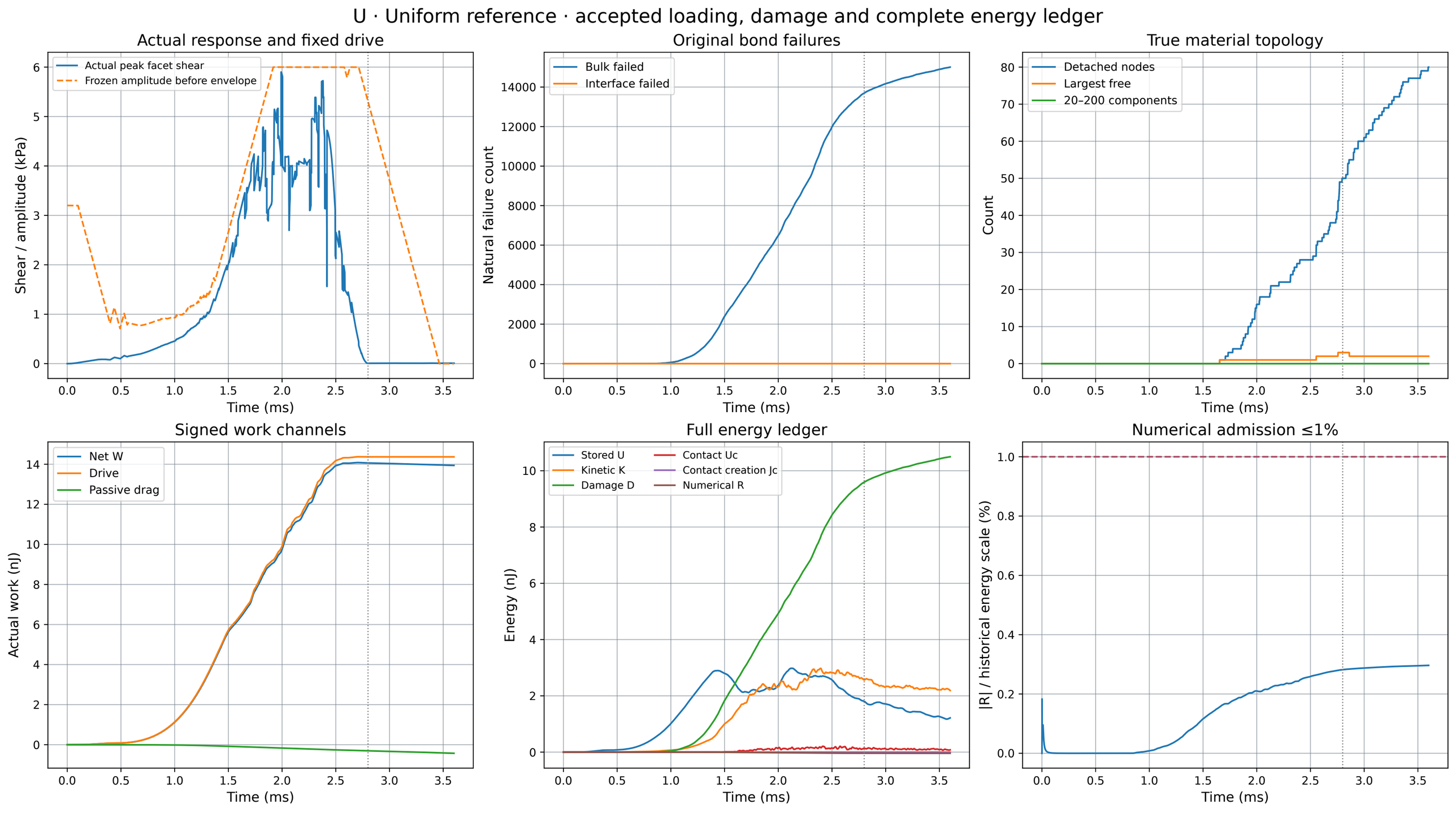

U · Uniform reference · accepted loading, damage and complete energy ledger
Actual response and fixed drive
Actual peak facet shear
Frozen amplitude before envelope
Shear / amplitude (kPa)
Time (ms)
Original bond failures
Bulk failed
Interface failed
Natural failure count
Time (ms)
True material topology
Detached nodes
Largest free
20–200 components
Count
Time (ms)
Signed work channels
Net W
Drive
Passive drag
Actual work (nJ)
Time (ms)
Full energy ledger
Stored U
Kinetic K
Damage D
Contact Uc
Contact creation Jc
Numerical R
Energy (nJ)
Time (ms)
Numerical admission ≤1%
|R| / historical energy scale (%)
Time (ms)